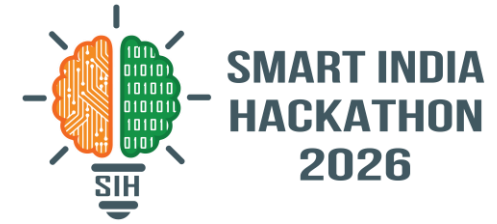


SMART INDIA HACKATHON 2026

TITLE PAGE



MOLE · Mine Observation & Live-alert Engine

Problem Statement ID SIH26025

Problem Statement Title Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India

Theme Disaster Management

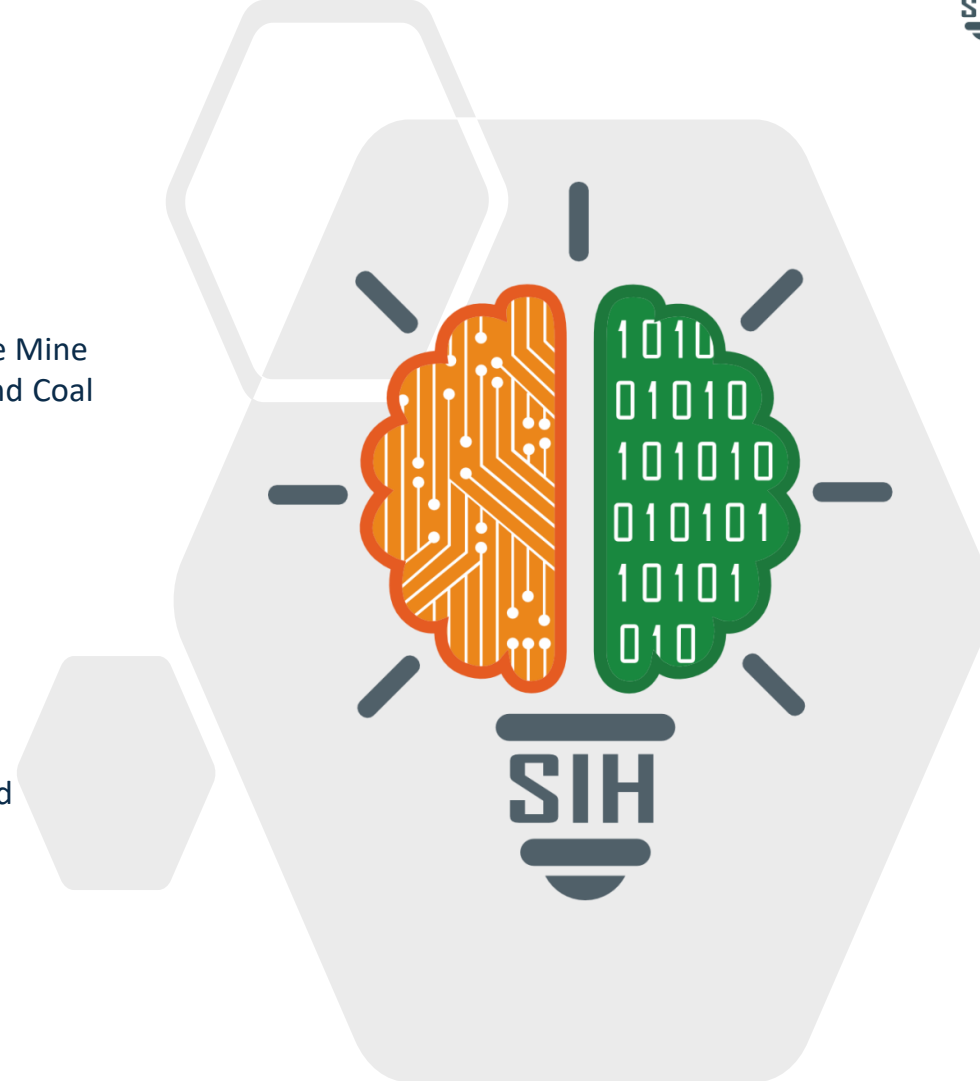
PS Category Hardware

Organisation Ministry of Coal

Team ID To be filled on the SIH portal after college shortlist

Team Name WE COOK

Idea MOLE — monitor the model, warn from evidence, inspect on command





MOLE — Proposed Solution



Two fixed nodes watch the model. Laptop rules + sklearn explain. Rover inspects only after the officer decides.

Proposed solution · one product, three jobs · rover is not the monitor



How this addresses SIH26025 · innovation = honest mapping, not a bigger claim

PS word	What MOLE actually does	What we do not claim
Monitoring	Live Node A + Node B, source-tagged and timed	Rover IMU is not Node B
Early warning	Rules: 3 samples → WATCH/ALERT; stale >5 s → UNKNOWN	Green = healthy data, not a certified-safe mine
AI-enabled	Isolation Forest + LOF vs this rig’s NORMAL; score + top feature	No LLM, no camera vision, no collapse %
Prediction	Holdout-selected Ridge: next 30 s of the same signals + MAE	Not a mine-subsidence or roof-fall forecast
Uniqueness	ML cannot clear a rule latch. A vs B residual. Rover only after a warning.	Not a gas-car maze robot; not LoRa; not autonomous

Technologies + method · two radios · one local laptop · rover is not step 6 of USB



NODES A + B

ESP32-WROOM-32 · MPU 0x68
A: slider GPIO34 B: no slider



RECEIVER

Waveshare ESP32-S3-Zero
ESP-NOW ch1 → USB JSON · no ML



LAPTOP ML

Flask · SQLite · scikit-learn
Isolation Forest + Ridge 30 s



ROVER

L298N · HC-SR04 · IR GPIO19
MQ-7 raw · AP Mine-Rover-AP

MONITOR PATH · both nodes broadcast ESP-NOW ch1 · USB into the laptop

Node A

MPU 21/22 · slider GPIO34

Node B

MPU 21/22 · compare only

S3-Zero USB

ESP-NOW RX
JSON 115200

Laptop

Validate + store
Rules · IF · Ridge

Dashboard

127.0.0.1 Flask
Reason · score · MAE

INSPECT PATH · separate Wi-Fi AP Mine-Rover-AP · not on the node USB radio

Officer

Sees WATCH / ALERT
Then decides to inspect

Laptop proxy

Hold FWD / REV / LEFT / RIGHT
Release sends STOP

Rover ESP32

L298N 13 / 12 / 14 / 27
Remote — not autonomous

Inspect sensors

HC-SR04 cm · IR flag
MQ-7 raw · rover IMU only

RULES (latch)

- 3 consecutive samples
- WATCH 3° or 2 mm
- ALERT 6° or 4 mm
- Stale >5 s → UNKNOWN

ISOLATION FOREST + LOF

- Unusual vs this rig's NORMAL
- Prior READY, or 120 live samples
- Score + top feature on screen
- Cannot clear a rule latch

30 s FORECAST

- Holdout-selected Ridge
- Next 30 s of same signals
- MAE shown on the chart
- Not a collapse prediction

We can run this on a table this week. It is not a real mine system.



Parts are with us

Node A, Node B, rover, USB board.



One laptop

Local website. No login. No cloud.



AI is ready

Loaded already. No long wait.



Short-range radio

Works across the table. Not through rock.

How we show it works (one loop on the table)

1 Still

Both nodes quiet

2 Tilt

Move Node A a little

3 Warn

WATCH then ALERT

4 Drive

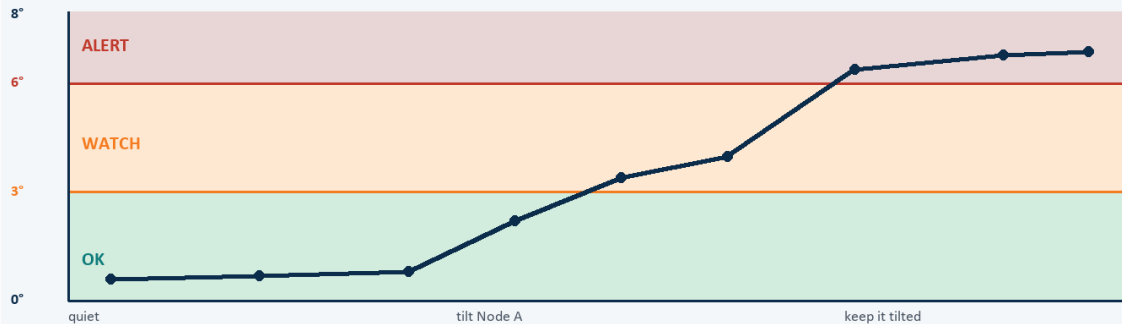
Open rover page

5 Keep

Event stays in history

Demo graph · when the light changes

Table model only. Not a mine safety chart.



Need 3 readings in a row. One tap is not a warning.
No packet for 5 s → UNKNOWN. Never fake a green light.
AI cannot turn a red warning back to green.



Cable dies

Show UNKNOWN. Never fake a green light.



Rover Wi-Fi dies

Motors stop on their own in ~0.4 s.



Easy to over-claim

No collapse %. Tilt is not ground sink.



Toy electronics

Table model only. Not mine-approved gear.



IMPACT AND BENEFITS



Target on demo day: a mine safety officer on a tabletop model. Ministry of Coal owns the PS. Impact = notice + evidence + remote look — not disaster-prevention numbers we cannot prove.

Who

Officer sees A vs B, rule reason, AI score, forecast MAE, then drives the rover because a warning appeared.

Social

Fewer unnecessary entries into the model tunnel. Decision stays human. Ack records awareness; it does not wipe history.

Economic / ops

Standing ESP32-class watch vs waiting for a specialist survey. One laptop site. USB still works if venue Wi-Fi is bad.

Environment / limit

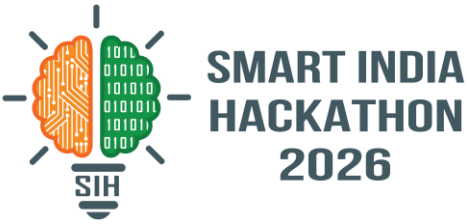
Hobby electronics on a model. Not intrinsically safe, not DGMS-approved, not a real underground vehicle.

Benefits a judge can verify on the table · nothing we invent

Benefit we can show	How a judge verifies it	Not a benefit we invent
Earlier notice of disturbance	Controlled tilt/slider → WATCH then latched ALERT	Lives saved / disasters averted
Evidence, not a vibe	Charts + A-vs-B + IF feature + Ridge MAE	87% collapse risk
Inspect without replacing monitors	Rover telemetry while A/B keep streaming	Autonomous rescue robot
Honest scale path	Same workflow later with surveyed mounts	This prototype already predicts subsidence



RESEARCH AND REFERENCES



References used to design the demo — not proof that MOLE is a certified mine instrument.

Item	Why it is here	Citation / link (no invented stats)
PS SIH26025	Official catalogue ID — never SIH2026025	SIH 2026 portal · Ministry of Coal · Hardware · Disaster Management
Isolation Forest	Anomaly vs this rig’s normal	Liu, Ting, Zhou — Isolation Forest, IEEE ICDM 2008
30 s forecast	Short-horizon prediction of measured signals	scikit-learn Ridge / holdout MAE on a recent window
MPU6050	Gravity tilt + RMS vibration proxy	InvenSense MPU-6000/6050 Product Specification
ESP-NOW + ADC1	Node radio ch1; slider on GPIO34	Espressif ESP-NOW User Guide; ESP32 ADC notes
ESP32-S3-Zero	USB CDC receiver, no sensors	Waveshare ESP32-S3-Zero wiki
MQ-7 / IR	Rover inspect only	MQ-7 datasheet — raw ADC, not ppm. IR = digital LOW flag, not ranging
Implementation	Partial working software + firmware	https://github.com/Rithvik-7/WECOOK-MOLE
Domain context only	Why mines care about movement	Coal Mines Regulations, 2017; DGMS (S&T) Circ. 01/2017 cl. 6.1 — not implemented here

Close: nodes detect → rules latch → Isolation Forest + Ridge explain → rover inspects.
Upload PDF of these 6 slides only. Team ID: fill on portal. No extra slide.